Open-World AI Image Detection with Abstention

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Abstract

Binary "AI vs. real" detectors often look strong on curated benchmarks but fail on the cases that matter. For example, images from unseen generators, reposting/JPEG compression, and adversarial perturbations. This project builds an **open-world** detector that outputs **AI, Real or Abstain** (instead of always guessing). The headline is reliability: calibrated confidence and risk–coverage curves showing that when the model chooses to answer, it is accurate. In this report, we discuss our current progress, including the challenges we faced and how we tackled them, along with our baseline results. Finally, we present our proposed next steps.

1 Introduction

Motivation. The generative landscape is non-stationary as new text-to-image models appear continuously, and detector performance often collapses under this open-world shift. For example, the Visual Counter Turing Test (VCT^2) [7] evaluates detectors on modern generators (e.g., Stable Diffusion (SD3/SD3.5) [4], DALL·E 3 [10], Midjourney 6 [8]) and reports that many detectors barely beat chance under unified settings [7]. Even if a detector is accurate on clean inputs, real-world transformations like JPEG compression and reposting can erase fragile forensic traces. Further, RAID (Real and AI-Generated Image Detection Dataset) [3] provides a large-scale stress test of transferable adversarial examples that can deceive state-of-the-art detectors.

Recent work. The work in Community Forensics [11] shows that training on thousands of generators improves generalization compared to small-generator datasets. VCT^2 [7] and RAID [3] provide modern and adversarial evaluations that expose the generalization and robustness gap [7, 3]. The Diffusion Reconstruction Error (DIRE) approach provides a diffusion-specific forensic representation that improves cross-diffusion generalization [13].

Goal. Our goal is to build a classification system that determines whether an image is AI-generated or real when the generator is unknown, and safely abstains when uncertain. More specifically, our detector predicts AI or Real only when confidence exceeds a threshold and abstains otherwise. We evaluate accuracy and calibration on the images it chooses to label, reflecting real-world settings where low-confidence guesses can be harmful.

2 Related Work

AI-image detection under distribution shift. Recent work shows that detector performance can degrade sharply when test data differs from training data in generator family, image pipeline, or post-processing [9, 3]. Community Forensics emphasizes scale and diversity in detector training, arguing that broad generator coverage improves generalization relative to small-generator settings [11]. However, benchmark studies such as VCT^2 highlight that many detectors remain brittle on newer

text-to-image systems and unified evaluation protocols [7]. This motivates open-world evaluation, in which the training and test distributions are intentionally mismatched.

Robustness and adversarial stress testing. Beyond natural shift, adversarial transfer further challenges detector reliability. RAID provides large-scale transferable attacks that can substantially reduce detection performance across models, exposing gaps between standard benchmark accuracy and real robustness [3]. These findings suggest that forced binary decisions are often unsafe in deployment settings where manipulations, reposting, and compression are common.

Forensic features and selective prediction. On the modeling side, Contrastive Language-Image Pretraining (CLIP) style representations provide strong generic visual features for lightweight downstream detectors [12], while DIRE proposes diffusion-reconstruction residual signals that can improve cross-diffusion forensic behavior [13]. On the uncertainty side, temperature scaling is a simple and effective post-hoc calibration method for reducing overconfidence [6]. Conformal prediction adds a complementary set-valued uncertainty layer with finite-sample coverage guarantees under exchangeability assumptions [2]. In selective classification, abstention is used to trade coverage for lower error on answered samples, typically summarized with risk–coverage curves and Area Under the Risk Curves (AURC) [5].

Positioning of this work. Our work combines these threads in a single open-world pipeline: a CLIP-based binary detector (with optional DIRE-style residual fusion), post-hoc calibration, and abstention via thresholding and split conformal prediction. The focus is not only on raw detection accuracy but also on reliability under shifts, measured by calibration quality and selective-performance metrics on the CommunityForensics, VCT^2, RAID, and ARIA benchmarks.

3 Method

3.1 Datasets

To train an image authenticity classifier, we require a large and diverse collection of both real and AI-generated images so the model can learn visual patterns that distinguish the two. Thus, we aim to train and calibrate our detector on CommunityForensics-Small [1]. which contains approximately 278K AI-generated images sampled from 4,803 different generator models, paired with 278K real images. Although the dataset provides binary labels (real vs. generated), our system will be designed to produce tri-state outputs by allowing abstention when confidence is low. For the open-world evaluation scenario, we aim to test our detector on data that differs from the training data in terms of generator type and image content. Our objective is to use VCT^2 to evaluate generalization to text-to-image models such as DALL·E 3 and Midjourney 6. We further aim to evaluate robustness using RAID, a set of transferable adversarial examples designed to fool state-of-the-art detectors, and ARIA, a diverse real-world dataset used to measure false positives under domain shift. Note that ARIA is not available on HuggingFace; we are currently exploring alternative access methods to incorporate it into our evaluation.

3.2 Baseline Detector

Our baseline detector builds on a pretrained CLIP vision encoder (ViT-B/32, OpenAI weights), which we use as a frozen feature extractor. For a given input image $x \in \mathbb{R}^{3\times224\times224}$, the encoder produces a 512-dimensional embedding $z = f_{\text{CLIP}}(x)$ that captures rich visual semantics without any task-specific fine-tuning. On top of this frozen backbone, we train a small MLP classification head with two hidden layers ($512 \to 256$ units), ReLU activations, and dropout ($p = 0.1$), ending in a two-class linear output layer. Only the head parameters are updated during training, keeping the CLIP encoder weights fixed throughout.

We train the head using cross-entropy loss with AdamW (lr $= 2 \times 10^{-4}$, weight decay $= 10^{-4}$) on CommunityForensics-Small. We reserve 10% of the training split as a held-out calibration set, which is used exclusively for temperature scaling and conformal threshold estimation after training is complete. At inference time, the logits are passed through a softmax to produce confidence scores, which feed directly into the abstention layer. Figure 1 shows the general process flow of our designed pipeline.

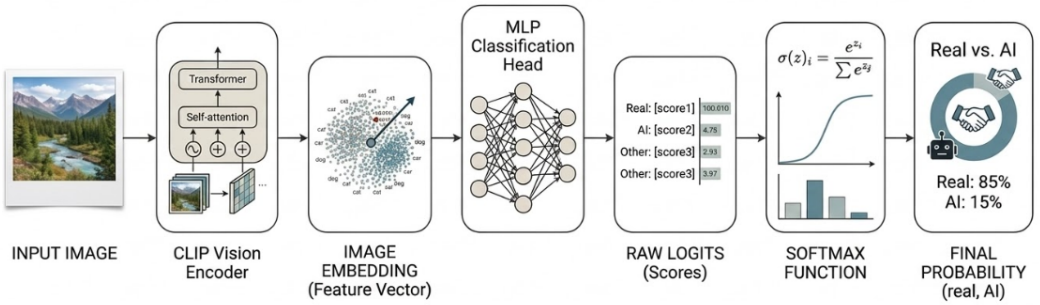

Figure 1: Overview of the general AI image generated detection pipeline.

3.3 Proposed DIRE-Based Forensic Fusion

While CLIP captures rich semantic features, it can overlook pixel-level artifacts that might differentiate AI-generated images from real images. Inspired by DIRE [13], we added a lightweight residual branch to complement the baseline model. Our core idea is simple: we smooth the input image using average pooling to get an approximation $\hat{x}$, then compute the residual $R = |x - \hat{x}|$. This difference map highlights fine-grained high-frequency details that the semantic features of CLIP tend to ignore.

We encode R through a single convolutional layer followed by adaptive average pooling into a compact 128-dimensional vector z_R. This is concatenated with the CLIP embedding z to form the fused representation $[z\,;\, z_R] \in \mathbb{R}^{640}$, which is then passed through the classification head. **This part is currently under work and we do not have results yet.**

4 Results

4.1 Evaluation method

The main success criterion of our work is that the system remains trustworthy under open-world shifts. The goal is to be accurate when the system gives an answer, and to abstain rather than make confident mistakes when the evidence is weak or degraded. Several metrics are to be used in order to evaluate our work. The Area Under the Receiver Operating Characteristic curve (AUROC) and True Positive Rate (TPR) at 1% False Positive Rate (FPR) are the standard detection quality in low-false-positive regimes. The Expected Calibration Error (ECE) is to be used, which checks whether "80% confidence" really means "correct 80% of the time" [6]. Furthermore, the risk coverage curve, along with AURC, helps in such applications as it plots the error rate against the coverage (the fraction of samples for which the model made predictions) and summarizes selective reliability. The lower AURC is better since it indicates that the model is achieving lower error while predicting larger fraction of input samples [5]. Finally, the abstention rate, which presents the abstain fraction and accuracy on the non-abstained subset (the main demo number).

4.2 Implemented Components

At the current stage, the core pipeline is implemented and runnable end-to-end. The baseline detector uses a pretrained CLIP vision encoder as a feature extractor with a lightweight MLP head for binary classification (Real vs. AI). The pipeline supports multiple evaluation modes: forced prediction and threshold-based abstention. For reliability-oriented inference, we added abstention mechanisms. Current abstentions is based on a confidence threshold τ, and we will implement a split conformal abstention mechanism which produces set-valued predictions and maps ambiguous sets to **Abstain**.

4.3 Experiments Run

Table 1: Baseline CLIP detector results on CommunityForensics and RAID test sets. Forced: all samples answered. Threshold ($\tau = 0.9$): abstain when max-softmax < 0.9.

Dataset	Mode	Abstain (%)	Acc (%)	AUROC	TPR@1%FPR	ECE	AURC
CommunityForensics	Forced	0.00	90.38	0.9630	0.7196	0.0414	0.0239
	Threshold	43.04	97.79	0.9630	0.7196	0.0414	0.1213
RAID	Forced	0.00	92.92	—	—	—	0.0096
	Threshold	45.90	99.82	—	—	—	0.1275

Using the implemented pipeline, we trained baseline model on CommunityForensics dataset with 96k samples. For the evaluations, we used 5k samples from CommunityForensics (evaluation set) and RAID. Table 1 reports the performance of the baseline CLIP detector under both forced prediction and threshold abstention.

When we used forced prediction, the model achieves 90.38% accuracy on CommunityForensics with an AUROC of 0.963. Our model showed good discriminative ability between real and AI-generated images. The TPR at 1% FPR of 71.96% indicates that the model can flag the majority of AI-generated images while keeping the false alarm rate very low. The ECE of 0.041 suggests the model is reasonably well calibrated out of the box.

On RAID, the model reaches 92.92% accuracy. Since RAID contains only AI-generated images, AUROC and TPR@1%FPR are not applicable, but the low AURC of 0.0096 confirms that the model assigns high confidence to its correct predictions consistently across the dataset.

If we apply a threshold abstention with $\tau = 0.9$, it improves selective accuracy substantially on both datasets. On CommunityForensics, accuracy rises from 90.38% to 97.79% while abstaining on 43% of samples. On RAID, selective accuracy reaches 99.82% with a 46% abstention rate. This shows that a simple confidence threshold can meaningfully filter out uncertain predictions, at the cost of leaving a significant fraction of inputs unanswered. The increase in AURC under threshold abstention reflects that the metric integrates over the full coverage range; concentrating answered samples at high-confidence regions improves accuracy but shifts risk to lower coverage values, raising the integral.

5 Ablation Study

At the current milestone of the project, we mainly focused on developing and evaluating the baseline CLIP-based detector. Since several components of our full pipeline are still under development, we have not yet conducted a complete ablation study. In the final version of the project, we plan to run ablation experiments to better understand the contribution of each component in the system. In particular, we will study the impact of the proposed residual forensic branch, the effect of calibration methods such as temperature scaling, and how different abstention thresholds influence the tradeoff between accuracy and coverage. In addition, we plan to experiment with fine-tuning the CLIP vision encoder on a subset of the CommunityForensics dataset to evaluate whether task-specific adaptation improves detection performance compared to using frozen CLIP encoder. These experiments will help us better understand the role of each component in the pipeline and guide improvements to the overall system.

6 Discussion & Conclusion

Discussion. Overall, the baseline training, calibration, and evaluation stack is functional. The model shows strong in-domain performance on CommunityForensics and RAID under forced and threshold prediction. This is consistent with prior work showing that large pretrained vision encoders provide powerful representations for downstream classification tasks. However, high in-domain accuracy alone does not guarantee reliability in practical settings. Previous studies have shown that AI-image detectors often fail when evaluated on unseen generators or under adversarial perturbations.

Therefore, the central question of this project is not whether the detector can achieve high accuracy on a curated dataset, but whether it remains trustworthy under distribution shift.

Abstention experiments highlight an important tradeoff between reliability and coverage. When the detector is allowed to abstain based on high confidence thresholds, the accuracy on answered samples becomes extremely high. However, this improvement is accompanied by a significant abstain rate, meaning that the model declines to answer on a large portion of inputs. This tradeoff is a core aspect of selective prediction systems. In practical deployment scenarios, such a tradeoff may still be desirable. A system that refuses to answer uncertain cases may be preferable to one that produces confident but incorrect predictions. Nevertheless, the abstain rate must be carefully controlled to ensure the system remains useful. The most important remaining evaluation is therefore cross-dataset testing under open-world conditions. If abstention significantly reduces errors on datasets such as VCT^2 while maintaining reasonable coverage, it would support the central claim that abstention improves reliability for AI-image detection.

Proposed next steps. The next phase of the project will focus on completing the open-world evaluation and refining the abstention mechanism. First, we will run full evaluations on the VCT^2, and ARIA datasets. These datasets introduce distribution shifts through unseen generators, adversarial examples, and real-world imagery. The goal is to compare three decision modes: forced prediction, threshold-based abstention, and conformal abstention. Second, we will study the tradeoff between accuracy and coverage by varying the abstention threshold and analyzing risk–coverage curves. This analysis will help determine whether selective prediction provides meaningful reliability improvements without excessively reducing coverage. Third, we will evaluate the optional DIRE-based forensic signal that incorporates diffusion reconstruction residuals. This component may help capture diffusion-specific artifacts that generalize better across diffusion generators. Finally, we will improve the efficiency and robustness of the data pipeline to support larger-scale experiments. This includes refining the streaming data loader and ensuring stable training across multiple large datasets. Since handling the large size of the benchmarks was the main challenge in implementing our baseline. Together, these steps will enable a comprehensive evaluation of the proposed open-world AI-image detection framework in the final report.

Deliverables. We aim to provide a demo that outputs AI, real, or abstain decisions with confidence scores for uploaded images, and a final comparison showing risk–coverage curves for a forced-decision baseline versus the abstaining system on VCT^2 and degraded inputs.

References

[1] Andrew Owens Research Lab. OwensLab/CommunityForensics-Small dataset card. `https://huggingface.co/datasets/OwensLab/CommunityForensics-Small`, 2025. Accessed 2026-02-03.

[2] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2021.

[3] Hicham Eddoubi, Jonas Ricker, Federico Cocchi, Lorenzo Baraldi, Angelo Sotgiu, Maura Pintor, Marcella Cornia, Asja Fischer, Rita Cucchiara, and Battista Biggio. Raid: A dataset for testing the adversarial robustness of ai-generated image detectors, 2025.

[4] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024.

[5] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks, 2017.

[6] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017. arXiv:1706.04599.

[7] Nasrin Imanpour, Abhilekh Borah, Shashwat Bajpai, Subhankar Ghosh, Sainath Reddy Sankepally, Hasnat Md Abdullah, Nishoak Kosaraju, Shreyas Dixit, Ashhar Aziz, Shwetangshu Biswas, Vinija Jain, Aman Chadha, Song Wang, Amit Sheth, and Amitava Das. The visual counter turing test (vct^2): A benchmark for evaluating ai-generated image detection and the visual ai index (vai), 2024. Last revised 2025.

[8] Midjourney, Inc. Midjourney. `https://www.midjourney.com`, 2023.

[9] Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. Towards universal fake image detectors that generalize across generative models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24480–24489, 2023.

[10] OpenAI. Dall·e 3: Improving image generation with better captions. `https://openai.com/research/dall-e-3`, 2023.

[11] Jeongsoo Park and Andrew Owens. Community forensics: Using thousands of generators to train fake image detectors. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 8245–8257, June 2025. arXiv:2411.04125.

[12] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021. arXiv:2103.00020.

[13] Zhendong Wang, Jianmin Bao, Wengang Zhou, Weilun Wang, Hezhen Hu, Hong Chen, and Houqiang Li. Dire for diffusion-generated image detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. arXiv:2303.09295.